LOWER BOUNDS FOR A POLYNOMIAL ON A BASIC CLOSED SEMIALGEBRAIC SET USING GEOMETRIC PROGRAMMING

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Abstract. Let $f, g_1, \ldots, g_m$ be elements of the polynomial ring $\mathbb{R}[x_1, \ldots, x_n]$. The paper deals with the general problem of computing a lower bound for f on the subset of $\mathbb{R}^n$ defined by the inequalities $g_i \geq 0$, $i = 1, \ldots, m$. The paper shows that there is an algorithm for computing such a lower bound, based on geometric programming, which applies in a large number of cases. The algorithm extends and generalizes earlier algorithms of Ghasemi and Marshall, dealing with the case $m = 0$, and of Ghasemi, Lasserre and Marshall, dealing with the case $m = 1$ and $g_1 = M - (x_1^d + \cdots + x_n^d)$. Here, d is required to be an even integer $d \geq \max\{2, \deg(f)\}$. The algorithm is implemented in a SAGE program developed by the first author. The bound obtained is typically not as good as the bound obtained using semidefinite programming, but it has the advantage that it is computable rapidly, even in cases where the bound obtained by semidefinite programming is not computable.

1. Introduction

Let $f, g_1, \ldots, g_m$ be elements of the polynomial ring $\mathbb{R}[\mathbf{x}] = \mathbb{R}[x_1, \cdots, x_n]$ and let

$$K_{\mathbf{g}} := \{\mathbf{x} \in \mathbb{R}^n : g_j(\mathbf{x}) \geq 0,\ j = 1, \ldots, m\}.$$

Here, $\mathbf{g} := (g_1, \ldots, g_m)$. We refer to $K_{\mathbf{g}}$ as the basic closed semialgebraic set generated by $\mathbf{g}$. Observe that if $m = 0$, then $\mathbf{g} = \emptyset$ and $K_{\mathbf{g}} = \mathbb{R}^n$. Let

$$f_{*,\mathbf{g}} := \inf\{f(\mathbf{x}) : \mathbf{x} \in K_{\mathbf{g}}\}.$$

One would like to have a simple algorithm for computing a lower bound for f on $K_{\mathbf{g}}$, i.e., a lower bound for $f_{*,\mathbf{g}}$. Lasserre's algorithm [6] is such an algorithm. It produces a hierarchy of lower bounds

$$f_{\mathrm{sos},\mathbf{g}}^{(t)} = \sup\{r \in \mathbb{R} : f - r = \sum_{j=0}^{m} \sigma_j g_j, \sigma_j \in \sum \mathbb{R}[\mathbf{x}]^2, \deg(\sigma_j g_j) \leq t, j = 0, \ldots, m\}$$

for f on $K_{\mathbf{g}}$, one for each integer $t \geq \max\{\deg(f), \deg(g_j) : j = 1, \ldots, m\}$, which are computable by semidefinite programming. Here, $g_0 := 1$ and $\sum \mathbb{R}[\mathbf{x}]^2$ denotes the set of elements of $\mathbb{R}[\mathbf{x}]$ which are sums of squares. Denote by d the least even integer $d \geq \max\{2, \deg(f), \deg(g_j) : j = 1, \ldots, m\}$. The algorithm in [5] deals

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with the case $m = 0$, producing a lower bound f_{gp} for f on $\mathbb{R}^n$ computable by geometric programming.[1] See [2], [4] and [7] for precursors of [5]. The algorithm in [3] is a variation of the algorithm in [5], which deals with the case $m = 1$, $g_1 = M - (x_1^d + \cdots + x_n^d)$, i.e., it produces a lower bound for f on the hyperellipsoid

$$B_M := \{\mathbf{x} \in \mathbb{R}^n : x_1^d + \cdots + x_n^d \leq M\}.$$

Again, this lower bound is computable by geometric programming. Of course, if $K_{\mathbf{g}}$ is compact, then $K_{\mathbf{g}} \subseteq B_M$ for M sufficiently large, so the lower bound established in [3] also provides a lower bound for f on $K_{\mathbf{g}}$.

Although the bounds obtained in [3] and [5] are typically not as good as the bounds obtained in [6], the computation is much faster, especially when the coefficients are sparse, and problems where the number of variables and the degree are large (problems where the method in [6] breaks down completely) can be handled easily.

The goal of the present paper is to establish a general lower bound for f on $K_{\mathbf{g}}$ computable by geometric programming. In case $m = 0$ it should be the lower bound f_{gp} obtained in [5]. In case $m = 1$ and $g_1 = M - (x_1^d + \cdots + x_n^d)$, it should be the lower bound obtained in [3]. This goal is not attained in every case, but it is attained in a large number of cases.

The idea is the following: Let $G(\lambda) = f - \sum_{j=1}^m \lambda_j g_j$ where $\lambda = (\lambda_1, \ldots, \lambda_m) \in [0, \infty)^m$. By [5] (also see Theorem 2.1 below), $G(\lambda)_{\mathrm{gp}}$ is a lower bound for $G(\lambda)$ on $\mathbb{R}^n$. It follows that $G(\lambda)_{\mathrm{gp}}$ is a lower bound for f on $K_{\mathbf{g}}$ and consequently, that

$$s(f, \mathbf{g}) := \sup\{G(\lambda)_{\mathrm{gp}} : \lambda \in [0, \infty)^m\}$$

is a lower bound for f on $K_{\mathbf{g}}$. By [5], for each $\lambda \in [0, \infty)^m$, $G(\lambda)_{\mathrm{gp}}$ is computable by geometric programming. Unfortunately, this does not imply that the supremum is so computable, although there are important cases where it is; see Theorem 4.2 (2). More to the point, there are important cases where, even though the supremum itself may not be computable by geometric programming, there is a relaxation which is computable by geometric programming; see Theorem 4.1 and Theorem 4.2 (1).

The main new result is Theorem 4.2. See Remark 4.3 (6) for the application of Theorem 4.2 to the computation of a lower bound on any product of hyperellipsoids. See Remark 4.3 (8), (9) and (10) for runtime and relative error computations. Theorem 5.1 explains how the hypothesis of Theorem 4.2 can be weakened slightly in the case $m = 1$. See Theorems 5.2 and 5.3 for other variants of Theorem 4.2.

[1] A function $\phi : (0, \infty)^n \to \mathbb{R}$ of the form $\phi(\underline{x}) = c x_1^{a_1} \cdots x_n^{a_n}$, where $c > 0$, $a_i \in \mathbb{R}$ and $\underline{x} = (x_1, \ldots, x_n)$ is called a *monomial function.* A sum of monomial functions is called a *posynomial function.* An optimization problem of the form

$$\begin{cases} \text{Minimize} & \phi_0(\underline{x}) \\ \text{Subject to} & \phi_i(\underline{x}) \leq 1,\ i = 1, \ldots, m \text{ and } \ \psi_j(\underline{x}) = 1,\ j = 1, \ldots, p \end{cases}$$

where $\phi_0, \ldots, \phi_m$ are posynomials and $\psi_1, \ldots, \psi_p$ are monomial functions, is called a *geometric program.* See [1] and [5].

See Remark 5.4 for some indication of how Theorems 4.2, 5.1, 5.2 and 5.3 can be applied in practice. See Example 5.5 for sample computations. Theorem 6.1 relates the lower bound on the hypercube $\prod_{j=1}^n[-N_j, N_j]$ described in Remark 4.3 (4) to the trivial lower bound introduced in [3, Section 3]. The source code of a SAGE program, developed by the first author, which computes the lower bound of f on $K_{\mathbf{g}}$ described in Theorem 4.1, is available at github.com/mghasemi/CvxAlgGeo.

2. THE CASE $m = 0$

We recall the algorithm established in [5]. We need some notation. Fix an even integer $d \geq 2$. Let $\mathbb{N}_d^n := \{\alpha \in \mathbb{N}^n : |\alpha| \leq d\}$ where $|\alpha| = \sum_i \alpha_i$ for every $\alpha \in \mathbb{N}^n$. Let $\epsilon_i := (\delta_{i1}, \cdots, \delta_{in}) \in \mathbb{N}^n$, with $\delta_{ij} = d$ if $i = j$ and 0 otherwise and, given $f = \sum f_\alpha \mathbf{x}^\alpha \in \mathbb{R}[\mathbf{x}]$, $\deg(f) \leq d$, let:

$$\begin{array}{lcl}
\Omega(f) & := & \{\alpha \in \mathbb{N}_d^n \,:\, f_\alpha \neq 0\} \setminus \{0, \epsilon_1, \cdots, \epsilon_n\} \\
\Delta(f) & := & \{\alpha \in \Omega(f) \,:\, f_\alpha\, \mathbf{x}^\alpha \text{ is not a square in } \mathbb{R}[\mathbf{x}]\} \\
\Delta(f)^{<d} & := & \{\alpha \in \Delta(f) \,:\, |\alpha| < d\} \\
\Delta(f)^{=d} & := & \{\alpha \in \Delta(f) \,:\, |\alpha| = d\}.
\end{array}$$

Denote the coefficient f_{ϵ_i} by $f_{d,i}$ for $i = 1, \ldots, n$. One is most interested in the case where $\deg(f) = d$.

Theorem 2.1. [5, Theorem 3.1] *Let $f \in \mathbb{R}[\mathbf{x}]$, $\deg(f) \leq d$, and let $\rho(f)$ denote the optimal value of the program:*

$$(1) \quad \begin{cases}
\textit{Minimize} & \displaystyle\sum_{\alpha \in \Delta(f)^{<d}} (d - |\alpha|) \left[\left(\frac{f_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha \right]^{1/(d-|\alpha|)} & \\
\textit{s.t.} & \displaystyle\sum_{\alpha \in \Delta(f)} z_{\alpha,i} \leq f_{d,i}, & i = 1, \ldots, n \\
& \left(\frac{\mathbf{z}_\alpha}{\alpha}\right)^\alpha = \left(\frac{f_\alpha}{d}\right)^d, & \alpha \in \Delta(f)^{=d}
\end{cases}$$

where, for every $\alpha \in \Delta(f)$, the unknowns $\mathbf{z}_\alpha = (z_{\alpha,i}) \in [0, \infty)^n$ satisfy $z_{\alpha,i} = 0$ if and only if $\alpha_i = 0$. Then $f - f(0) + \rho(f)$ is a sum of binomial squares. In particular, $f_{\mathrm{gp}} := f(0) - \rho(f)$ is a lower bound for f on $\mathbb{R}^n$.

Here, $\left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha := \prod_{i=1}^n \frac{\alpha_i^{\alpha_i}}{(z_{\alpha,i})^{\alpha_i}}$ and $\left(\frac{\mathbf{z}_\alpha}{\alpha}\right)^\alpha := \prod_{i=1}^n \frac{(z_{\alpha,i})^{\alpha_i}}{\alpha_i^{\alpha_i}}$, the convention being that $0^0 = 1$.

In (1), the constraint $\left(\frac{\mathbf{z}_\alpha}{\alpha}\right)^\alpha = \left(\frac{f_\alpha}{d}\right)^d$ can be replaced by the weaker constraint $\left(\frac{\mathbf{z}_\alpha}{\alpha}\right)^\alpha \geq \left(\frac{f_\alpha}{d}\right)^d$, for each $\alpha \in \Delta(f)^{=d}$. If $\mathbf{z}$ is a feasible point for the latter program then, by shrinking suitably the $z_{\alpha,i}$, $\alpha \in \Delta(f)^{=d}$, one gets a feasible point $\mathbf{z}'$ for the former program such that the objective function of (1) evaluated at $\mathbf{z}$ and $\mathbf{z}'$ are the same.

If the feasible set of the program (1) is empty, then $\rho(f) = \infty$ and $f_{\mathrm{gp}} = -\infty$. A sufficient (but not necessary) condition for the feasible set of (1) to be nonempty

is that $\Delta(f)^{=d} = \emptyset$ and $f_{d,i} > 0$, $i = 1, \dots, n$. If $\deg(f) < d$ then either $\Delta(f) = \emptyset$ and $f_{\mathrm{gp}} = f(0)$ or $\Delta(f) \neq \emptyset$ and $f_{\mathrm{gp}} = -\infty$.

If $f_{d,i} > 0$, $i = 1, \dots, n$ then (1) is a geometric program. Somewhat more generally, if $\forall\ i = 1, \dots, n$ either $(f_{d,i} > 0)$ or $(f_{d,i} = 0$ and $\alpha_i = 0\ \forall\ \alpha \in \Delta(f))$, then (1) is a geometric program. In the remaining cases (1) is not a geometric program and the feasible set of (1) is empty.

3. General case

We return to the set-up considered in the introduction, i.e., $f, g_1, \dots, g_m \in \mathbb{R}[\mathbf{x}]$, $K_{\mathbf{g}} = \{\mathbf{x} \in \mathbb{R}^n : g_j(\mathbf{x}) \geq 0, j = 1, \dots, m\}$. We denote by d the least even integer $d \geq \max\{2, \deg(f), \deg(g_j) : j = 1, \dots, m\}$. We define $G(\lambda) = f - \sum_{j=1}^m \lambda_j g_j$, $\lambda = (\lambda_1, \dots, \lambda_m) \in [0, \infty)^m$. Note that $G(\lambda)_\alpha = f_\alpha - \sum_{j=1}^m \lambda_j (g_j)_\alpha$, $G(\lambda)_{d,i} = f_{d,i} - \sum_{j=1}^m \lambda_j (g_j)_{d,i}$, and $G(\lambda)(0) = f(0) - \sum_{j=1}^m \lambda_j g_j(0)$. As explained already, $s(f, \mathbf{g}) := \sup\{G(\lambda)_{\mathrm{gp}} : \lambda \in [0, \infty)^m\}$ is a lower bound for f on $K_{\mathbf{g}}$.[2] Let $\Delta := \Delta(f) \cup \Delta(-g_1) \cup \cdots \cup \Delta(-g_m)$, $\Delta^{<d} := \{\alpha \in \Delta : |\alpha| < d\}$, $\Delta^{=d} := \{\alpha \in \Delta : |\alpha| = d\}$.

It is convenient to define $g_0 := -f$, $\lambda_0 := 1$, so $G(\lambda) = -\sum_{j=0}^m \lambda_j g_j$. *We also assume from now on that* $\Omega(-g_j) = \Delta(-g_j)$ *for each* $j = 0, \dots, m$. One can reduce to this case by ignoring all terms corresponding to elements of $\Omega(-g_j) \backslash \Delta(-g_j)$, i.e., by replacing g_j by

$$g_j' := g_j(0) + \sum_{\alpha \in \Delta(-g_j)} (g_j)_\alpha \mathbf{x}^\alpha + \sum_{i=1}^n (g_j)_{d,i} x_i^d,\ j = 0, \dots, m.$$

Then $-g_j' \leq -g_j$ on $\mathbb{R}^n$, $j = 0, \dots, m$, so $K_{\mathbf{g}} \subseteq K_{\mathbf{g}'}$ where $\mathbf{g}' := (g_1', \dots, g_m')$ and the minimum of $-g_0'$ on $K_{\mathbf{g}'}$ is not greater than the minimum of $-g_0$ on $K_{\mathbf{g}}$.

We consider the following program:

$$(2) \qquad \begin{cases} \text{Minimize} & \sum\limits_{j=1}^m \lambda_j g_j(0) + \sum\limits_{\alpha \in \Delta^{<d}} (d - |\alpha|) \left[\left(\frac{G(\lambda)_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)} \\ \text{s.t.} & \sum\limits_{\alpha \in \Delta} z_{\alpha,i} \leq G(\lambda)_{d,i}, \qquad i = 1, \dots, n \\ & \left(\frac{\mathbf{z}_\alpha}{\alpha} \right)^\alpha \geq \left(\frac{G(\lambda)_\alpha}{d} \right)^d, \qquad \alpha \in \Delta^{=d} \end{cases}$$

where, for every $\alpha \in \Delta$, the unknowns $\mathbf{z}_\alpha = (z_{\alpha,i}) \in [0, \infty)^n$ satisfy $z_{\alpha,i} = 0$ if and only if $\alpha_i = 0$, and the unknowns $\lambda = (\lambda_1, \dots, \lambda_m)$ satisfy $\lambda_j \geq 0$.

Theorem 3.1. *Denote by ρ the optimum value of (2). Then $f(0) - \rho$ is a lower bound for $s(f, \mathbf{g})$.*

Proof. Note that $\Delta(G(\lambda)) \subseteq \Delta$ for each $\lambda \in [0, \infty)^m$. For suppose $\alpha \in \Delta(G(\lambda))$. If $2 \nmid \alpha$ then $G(\lambda)_\alpha \neq 0$, so $f_\alpha \neq 0$ or $(g_j)_\alpha \neq 0$ for some j. If $2 \mid \alpha$ then $G(\lambda)_\alpha < 0$

[2] In fact, $s(f, \mathbf{g}) \leq f_{\mathrm{sos},\mathbf{g}}^{(d)}$. By Theorem 2.1, $G(\lambda) - G(\lambda)_{\mathrm{gp}}$ is a sum of binomial squares (obviously of degree at most d) for each $\lambda \in [0, \infty)^m$. This implies that $G(\lambda)_{\mathrm{gp}} \leq f_{\mathrm{sos},\mathbf{g}}^{(d)}$ for each $\lambda \in [0, \infty)^m$.

so $f_\alpha < 0$ or $(g_j)_\alpha > 0$ for some j. Note also that if $(\mathbf{z}, \lambda)$ is a feasible point of (2), then

$$\sum_{\alpha \in \Delta^{<d}} (d - |\alpha|) \left[\left(\frac{G(\lambda)_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)}$$

is an upper bound for $\rho(G(\lambda))$, so

$$f(0) - \sum_{j=1}^m \lambda_j g_j(0) - \sum_{\alpha \in \Delta^{<d}} (d - |\alpha|) \left[\left(\frac{G(\lambda)_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)}$$

is a lower bound for $G(\lambda)_{\mathrm{gp}} = G(\lambda)(0) - \rho(G(\lambda))$, for each feasible point $(\mathbf{z}, \lambda)$ of (2). It follows that $f(0) - \rho$ is a lower bound for $s(f, \mathbf{g})$. □

A sufficient (but not necessary) condition for the feasible set of (2) to be nonempty is that $\Delta^{=d} = \emptyset$ and there exists $\lambda \in [0, \infty)^m$ such that $G(\lambda)_{d,i} > 0$, $i = 1, \ldots, n$.

Unfortunately, (2) is generally not a geometric program, even if one replaces the constraint $\lambda_j \geq 0$ by $\lambda_j > 0$, for $j = 1, \ldots, m$.[3] Note also that $f(0) - \rho$ may be strictly smaller than $s(f, \mathbf{g})$.

Example 3.2. Suppose $n = 2$, $m = 1$, $f = x^2 - 2xy + y^2$, $g_1 = x + y$. Then $G(0)_{\mathrm{gp}} = f_{\mathrm{gp}} = 0$, $G(\lambda)_{\mathrm{gp}} = -\infty$ for $\lambda > 0$, so $s(f, \mathbf{g}) = G(0)_{\mathrm{gp}} = 0$. In this example, $f(0) - \rho = -\infty$. Similarly, if $f = x + y + x^2 - 2xy + y^2$, $g_1 = x + y$, then $G(1)_{\mathrm{gp}} = 0$, $G(\lambda) = -\infty$ for $\lambda \geq 0$, $\lambda \neq 1$, $s(f, \mathbf{g}) = G(1)_{\mathrm{gp}} = 0$, and $f(0) - \rho = -\infty$.

4. Relaxation to a geometric program

We discuss relaxations of (2) which are geometric programs. We consider a linear change of variables

$$\lambda_j = \sum_{k=0}^m a_{jk} \mu_k, \ j = 0, \ldots, m,$$

where $\mu_0 := 1$ and a_{jk}, $j, k = 0, \ldots, m$ are real constants such that

$$a_{0k} = \begin{cases} 1 \text{ if } k = 0 \\ 0 \text{ if } k \neq 0. \end{cases}$$

Let $h_k := \sum_{j=0}^m a_{jk} g_j$, $k = 0, \ldots, m$, $H(\mu) := -\sum_{j=0}^m \mu_j h_j$. Clearly $G(\lambda) = H(\mu)$. For $\alpha \in \Delta$, decompose $H(\mu)_\alpha$ as $H(\mu)_\alpha = H(\mu)^+_\alpha - H(\mu)^-_\alpha$, where

$$H(\mu)^+_\alpha := -\sum_{(h_j)_\alpha < 0} (h_j)_\alpha \mu_j, \ H(\mu)^-_\alpha := \sum_{(h_j)_\alpha > 0} (h_j)_\alpha \mu_j.$$

[3] If one replaces the constraints $\lambda_j \geq 0$ by $\lambda_j > 0$, for $j = 1, \ldots, m$, then (2) can be seen as a signomial geometric program. See [1, Section 9.1] for the definition of a signomial geometric program. Unfortunately, signomial geometric programs are non-convex and are typically much harder to solve than geometric programs.

We take advantage of the inequality $\max\{a,b\} \geq |a-b|$, which holds for any nonnegative real numbers a,b. Note that $\max\{a,b\} = |a-b|$ if and only if one of a,b is zero. Define $h_j(0)^+ := \max\{h_j(0),0\}$. We consider the following program:

$$
(3) \quad \begin{cases}
\text{Minimize} & \sum_{j=1}^{m} \mu_j h_j(0)^+ + \sum_{\alpha\in\Delta^{<d}} (d-|\alpha|)\left[\left(\frac{w_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)} & \\
\text{s.t.} & \sum_{\alpha\in\Delta} z_{\alpha,i} \leq H(\mu)_{d,i}, & i=1,\dots,n \\
& \left(\frac{\mathbf{z}_\alpha}{\alpha}\right)^\alpha \geq \left(\frac{w_\alpha}{d}\right)^d, & \alpha\in\Delta^{=d} \\
& w_\alpha \geq \max\{H(\mu)^+_\alpha, H(\mu)^-_\alpha\}, & \alpha\in\Delta \\
& \sum_{k=0}^{m} a_{jk}\mu_k \geq 0, & j=1,\dots,m
\end{cases}
$$

where, for every $\alpha\in\Delta$, the unknowns $\mathbf{z}_\alpha = (z_{\alpha,i}) \in [0,\infty)^n$ satisfy $z_{\alpha,i}=0$ if and only if $\alpha_i = 0$, the unknowns $\mathbf{w} = (w_\alpha)_{\alpha\in\Delta}$ satisfy $w_\alpha > 0$, and the unknowns $\mu = (\mu_1,\dots,\mu_m)$ satisfy $\mu_k > 0$.

Theorem 4.1. *Assume that exactly one of $a_{j0},\dots,a_{jm}$ is strictly positive, or all of $a_{j0},\dots,a_{jm}$ are non-negative, for each $j=1,\dots,m$, and exactly one of $(h_0)_{d,i},\dots,(h_m)_{d,i}$ is strictly negative, for each $i=1,\dots,n$. Then (3) is a geometric program. Moreover, if ρ denotes the optimum value of (3), then $f_{\mathrm{gp},\mathbf{g}} := -h_0(0)-\rho$ is a lower bound for $s(f,\mathbf{g})$.*

Proof. The constraint $\sum_{\alpha\in\Delta} z_{\alpha,i} \leq H(\mu)_{d,i}$ can be written in the form $\sum_{\alpha\in\Delta} z_{\alpha,i} + \sum_{j\neq j_i}(h_j)_{d,i}\mu_j \leq -(h_{j_i})_{d,i}\mu_{j_i}$, where j_i is the unique j such that $(h_j)_{d,i} < 0$. If exactly one of $a_{j0},\dots,a_{jm}$ is strictly positive, then the constraint $\sum_{k=0}^m a_{jk}\mu_k \geq 0$ can be written in the form $-\sum_{k\neq k_j} a_{jk}\mu_k \leq a_{jk_j}\mu_{k_j}$ where k_j is the unique k such that $a_{jk} > 0$. If all of $a_{j0},\dots,a_{jm}$ are non-negative, then the constraint $\sum_{k=0}^m a_{jk}\mu_k \geq 0$ is the empty constraint. Also, for each $\alpha\in\Delta$, $H(\mu)^+_\alpha$ and $H(\mu)^-_\alpha$ are posinomials in the μ_j and, for each $j=1,\dots,m$, $h_j(0)^+ \geq 0$. It follows from these facts that (3) is a geometric program. Suppose now that $(\mathbf{z},\mathbf{w},\mu)$ is a feasible point for (3). Let $\lambda_j = \sum_{k=0}^m a_{jk}\mu_k$, $j=0,\dots,m$. Then $\lambda_j \geq 0$, $j=1,\dots,m$, and $H(\mu) = G(\lambda)$. Also, $w_\alpha \geq \max\{H(\mu)^+_\alpha, H(\mu)^-_\alpha\} \geq |H(\mu)_\alpha| = |G(\lambda)_\alpha|$, for each $\alpha\in\Delta$, so $(\mathbf{z},\lambda)$ is a feasible point of (2). Also,

$$
\begin{aligned}
h_0(0) \quad + \quad & \sum_{j=1}^{m} \mu_j h_j(0)^+ + \sum_{\alpha\in\Delta^{<d}} (d-|\alpha|)\left[\left(\frac{w_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)} \\
\geq \quad & \sum_{j=0}^{m} \mu_j h_j(0) + \sum_{\alpha\in\Delta^{<d}} (d-|\alpha|)\left[\left(\frac{w_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)} \\
\geq \quad & \sum_{j=0}^{m} \mu_j h_j(0) + \sum_{\alpha\in\Delta^{<d}} (d-|\alpha|)\left[\left(\frac{H(\mu)_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)} \\
= \quad & \sum_{j=0}^{m} \lambda_j g_j(0) + \sum_{\alpha\in\Delta^{<d}} (d-|\alpha|)\left[\left(\frac{G(\lambda)_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)},
\end{aligned}
$$

so, by Theorem 3.1,

$$-h_0(0) - \sum_{j=1}^{m} \mu_j h_j(0)^+ - \sum_{\alpha \in \Delta^{<d}} (d - |\alpha|) \left[\left(\frac{w_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)}$$

is a lower bound for $s(f, \mathbf{g})$. Since this is valid for any feasible point $(\mathbf{z}, \mathbf{w}, \mu)$ of (3), it follows that $-h_0(0) - \rho$ is a lower bound for $s(f, \mathbf{g})$. □

One would expect the bound $f_{\mathrm{gp},\mathbf{g}}$ given by Theorem 4.1 to be best when $h_j(0) \geq 0$ for all $j = 1, \ldots, m$ and one of $H(\mu)^+_\alpha$, $H(\mu)^-_\alpha$ is identically zero for almost all $\alpha \in \Delta$. Note that one of $H(\mu)^+_\alpha$, $H(\mu)^-_\alpha$ is zero if and only if $\max\{H(\mu)^+_\alpha, H(\mu)^-_\alpha\} = |H(\mu)|$ if and only if the $(h_0)_\alpha, \ldots, (h_m)_\alpha$ are all greater than or equal to 0 or all less than or equal to 0.

The source code of a SAGE program, which outputs the lower bound $f_{\mathrm{gp},\mathbf{g}}$ of f on $K_\mathbf{g}$ described in Theorem 4.1 (or the statement "not a geometric program" if the hypothesis of Theorem 4.1 fails to hold), is available at the address github.com/mghasemi/CvxAlgGeo. The input is the polynomials $f, g_1, \ldots, g_m$, and the matrix $A = (a_{jk})$.

It is important to understand that $f_{\mathrm{gp},\mathbf{g}}$ depends on the choice of A. We write $f_{\mathrm{gp},\mathbf{g}} = f^A_{\mathrm{gp},\mathbf{g}}$ when we wish to emphasis this fact. A deficiency in Theorem 4.1 is that there is no indication of how the matrix A should be chosen. We proceed to address this deficiency now, in an important special case.

Theorem 4.2. *Assume $(*)$: for each $i = 1, \ldots, n$ there exists $0 \leq j_i \leq m$ such that $(g_{j_i})_{d,i} < 0$ and $(g_j)_{d,i} = 0$ for $j > j_i$. Then (1) there exists a canonically defined lower triangular matrix $A = (a_{jk})$ such that $a_{jj} = 1$, $a_{jk} = 0$ if $k > j$ and $a_{jk} \leq 0$ if $k < j$, such that the hypothesis of Theorem 4.1 holds for A, and (2) if, in addition, $\Delta(-g_j) = \emptyset$ for $j = 1, \ldots, m$, and $g_k(0) + \sum_{j>k} a_{jk} g_j(0) \geq 0$, for all $k = 1, \ldots, m$, then $f^A_{\mathrm{gp},\mathbf{g}} = s(f, \mathbf{g})$.*

Proof. Assume that $(*)$ holds. We know that $h_k = \sum_{j=0}^m a_{jk} g_j$, $k = 0, \ldots, m$. Choose $a_{jj} = 1$ and $a_{jk} = 0$ if $k > j$. Thus $h_k = g_k + \sum_{j>k} a_{jk} g_j$. For each $k < j$ define a_{jk} by induction on j, as follows. For each $k < j$ and each i such that $j = j_i$, choose $a_{jk} \leq 0$ as large as possible in absolute value so that $(h_k)_{d,i} = (g_k)_{d,i} + \sum_{j \geq j' > k} a_{j'k} (g_{j'})_{d,i} \geq 0$, i.e.,

$$a_{jk} := \min_{i=1,\ldots,n} \{-[(g_k)_{d,i} + \sum_{j>j'>k} a_{j'k} (g_{j'})_{d,i}]/(g_j)_{d,i}, 0 : j = j_i\}.$$

Note that $a_{j'k}$ is already defined, by induction on j, for $j > j' > k$. By choice of a_{jk}, for each i, if $j = j_i$, then $(h_k)_{d,i} = 0$ for $k > j$, $(h_j)_{d,i} = (g_j)_{d,i} < 0$, and $(h_k)_{d,i} \geq 0$ for $k < j$. It follows that the hypothesis of Theorem 4.1 holds.

Assume now that $\Delta(-g_j) = \emptyset$ for $j = 1, \ldots, m$, and $g_k(0) + \sum_{j>k} a_{jk} g_j(0) \geq 0$, for all $k = 1, \ldots, m$. We want to show $f^A_{\mathrm{gp},\mathbf{g}} = s(f, \mathbf{g})$. In view of Theorem 4.1 it suffices to show $f^A_{\mathrm{gp},\mathbf{g}} \geq s(f, \mathbf{g})$. By our hypothesis, $h_k(0) = g_k(0) + \sum_{j>k} a_{jk} g_j(0) \geq 0$,

for all $k = 1, \dots, m$, and $(g_j)_\alpha = 0$ for all $\alpha \in \Delta$ and all $j = 1, \dots, m$. Consequently, $H(\mu)_\alpha = G(\lambda)_\alpha = f_\alpha$ and $\max\{H(\mu)^+_\alpha, H(\mu)^-_\alpha\} = |f_\alpha|$, for all $\alpha \in \Delta$. Let $\lambda \in [0, \infty)^m$ and let $\epsilon > 0$ be given and let $\mathbf{z}$ be a feasible point of the the program

$$
(4) \qquad \begin{cases} \text{Minimize} & \displaystyle\sum_{\alpha \in \Delta(f)^{<d}} (d - |\alpha|) \left[\left(\frac{f_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)} \\ \text{s.t.} & \displaystyle\sum_{\alpha \in \Delta(f)} z_{\alpha,i} \le G(\lambda)_{d,i}, \qquad i = 1, \dots, n \\ & \left(\frac{\mathbf{z}_\alpha}{\alpha} \right)^\alpha = \left(\frac{f_\alpha}{d} \right)^d, \qquad \alpha \in \Delta(f)^{=d} \end{cases}
$$

Choose $\lambda' \in [0, \infty)^m$ so that $\lambda'_j > \lambda_j$, $j = 1, \dots, m$ and $G(\lambda)_{d,i} \le G(\lambda')_{d,i}$, $i = 1, \dots, n$ with λ' so close to λ that $|\sum_{j=1}^m (\lambda'_j - \lambda_j) g_j(0)| < \epsilon$. Existence of λ' is a consequence of $(*)$. For each $i = 1, \dots, n$ there exists $0 \le k (= j_i) \le m$ such that $(g_k)_{d,i} < 0$ and $(g_j)_{d,i} = 0$ for $j > k$, so

$$
\begin{aligned}
G(\lambda')_{d,i} - G(\lambda)_{d,i} &= -\sum_{j=0}^m (\lambda'_j - \lambda_j)(g_j)_{d,i} \\
&= -\sum_{j=0}^k (\lambda'_j - \lambda_j)(g_j)_{d,i} \\
&\ge -(\lambda'_k - \lambda_k)(g_k)_{d,i} - \sum_{j=0}^{k-1} (\lambda'_j - \lambda_j)|(g_j)_{d,i}|.
\end{aligned}
$$

Thus we can choose $\lambda'_1, \dots, \lambda'_m$ so that $\lambda'_j > \lambda_j$, $j = 1, \dots, m$, $|\sum_{j=1}^m (\lambda'_j - \lambda_j) g_j(0)| < \epsilon$, and so that if $k = j_i$, then $(\lambda'_j - \lambda_j)|(g_j)_{d,i}| \le -\frac{1}{2^{k-j}} (\lambda'_k - \lambda_k)(g_k)_{d,i}$, for $j < k$, so $-(\lambda'_k - \lambda_k)(g_k)_{d,i} \ge \sum_{j=0}^{k-1} (\lambda'_j - \lambda_j)|(g_j)_{d,i}|$. Choose $\mu' \in [0, \infty)^m$ so that $\lambda'_j = \sum_{k \le j} a_{kj} \mu'_k$, $j = 0, \dots, m$, $\lambda'_0 = \mu'_0 = 1$. Using $a_{jk} \le 0$ for $j > k$ and $\lambda'_j > 0$ one sees that $\mu'_j > 0$, $j = 1, \dots, m$. Finally, let $\mathbf{w}_\alpha = |f_\alpha|$ for all $\alpha \in \Delta$. One checks that $(\mathbf{z}, \mathbf{w}, \mu')$ is a feasible point for program (3) and

$$
\begin{aligned}
H(\mu')(0) \quad - & \sum_{\alpha \in \Delta(f)^{<d}} (d - |\alpha|) \left[\left(\frac{\mathbf{w}_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)} \\
= \; & G(\lambda')(0) - \sum_{\alpha \in \Delta(f)^{<d}} (d - |\alpha|) \left[\left(\frac{f_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)} \\
\ge \; & G(\lambda)(0) - \sum_{\alpha \in \Delta(f)^{<d}} (d - |\alpha|) \left[\left(\frac{f_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)} - \epsilon.
\end{aligned}
$$

It follows that $f^A_{\mathrm{gp},\mathbf{g}} \ge G(\lambda)_{\mathrm{gp}} - \epsilon$. □

Remark 4.3.

(1) If $(*)$ holds, $\Delta(-g_j) = \emptyset$ for $j = 1, \dots, m$, and the matrix A is chosen as in Theorem 4.2, then $H(\mu)_\alpha = f_\alpha$, for all $\alpha \in \Delta$, and program (3) reduces to the

following one:

$$
(5)\qquad \begin{cases} \text{Minimize} & \sum_{j=1}^{m} \mu_j h_j(0)^+ + \sum_{\alpha\in\Delta^{<d}} (d-|\alpha|)\left[\left(\frac{f_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)} \\ \text{s.t.} & \sum_{\alpha\in\Delta} z_{\alpha,i} \le H(\mu)_{d,i}, \quad i=1,\dots,n \\ & \left(\frac{\mathbf{z}_\alpha}{\alpha}\right)^\alpha = \left(\frac{f_\alpha}{d}\right)^d, \qquad \alpha\in\Delta^{=d} \\ & \sum_{k=0}^{m} a_{jk}\mu_k \ge 0, \qquad j=1,\dots,m \end{cases}
$$

where, for every $\alpha \in \Delta$, the unknowns $\mathbf{z}_\alpha = (z_{\alpha,i}) \in [0,\infty)^n$ satisfy $z_{\alpha,i} = 0$ if and only if $\alpha_i = 0$, and the unknowns $\mu = (\mu_1,\dots,\mu_m)$ satisfy $\mu_k > 0$.

(2) If condition $(*)$ is replaced by the stronger condition $(**)$: for each $i = 1,\dots,n$ there exists k such that $(g_k)_{d,i} < 0$, $(g_j)_{d,i} = 0$ for $j > k$ and $(g_j)_{d,i} \ge 0$ for $0 < j < k$, then $a_{jk} = 0$ for all $j > k > 0$, $h_k(0) = g_k(0) + \sum_{j>k} a_{jk} g_j(0) = g_k(0)$, and the condition $g_k(0) + \sum_{j>k} a_{jk} g_j(0) \ge 0$ reduces to $g_k(0) \ge 0$, $k = 1,\dots,m$.

(3) If $m = 1$ and $g_1 = M - (x_1^d + \cdots + x_n^d)$, then parts (1) and (2) of Theorem 4.2 apply, yielding a lower bound for f on the hyperellipsoid $B_M = \{\mathbf{x} \in \mathbb{R}^n : x_1^d + \cdots + x_n^d \le M\}$, computable by geometric programming. Note that, in this example,

$$
A = \begin{pmatrix} 1 & 0 \\ -c & 1 \end{pmatrix}, \text{ where } c = \max\{0, f_{d,i} : i = 1,\dots,n\}.
$$

Note also that, in this example, the program (5) for computing $f^A_{\mathrm{gp},\mathbf{g}}$ is not the same as the program in [3, Theorem 2.4], but it is equivalent, i.e., it produces the same output. In fact, because fewer variables and constraints are involved, the program (5) is faster than the one in [3]. These facts were also verified experimentally, by redoing Examples 4.1–4.5 of [3] using program (5) in place of the program used in [3].

(4) Fix $N_i > 0$, $i = 1,\dots,n$. If $m = n$, $g_i = N_i^d - x_i^d$, $i = 1,\dots,n$, then parts (1) and (2) of Theorem 4.2 apply. This gives a lower bound for f on the hypercube $\prod_{i=1}^n [-N_i, N_i]$. Observe that, in this example, the stronger condition $(**)$ holds, and

$$
A = \begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ -c_1 & 1 & \dots & 0 & 0 \\ & & \dots & & \\ -c_n & 0 & \dots & 0 & 1 \end{pmatrix} \text{ where } c_j = \max\{0, f_{d,j}\},\ j = 1,\dots,n
$$

(5) In computing a lower bound for f on $\prod_{j=1}^n [-N_j, N_j]$ using the method described in (4) one can take d to be any even integer $\ge 2\lceil \frac{\deg(f)}{2} \rceil$. We will show later, in Section 5, that if $d > \deg f$, more generally, if $d \ge \deg f$ and $f_{d,i} \le 0$, $i = 1,\dots,n$, then the lower bound obtained in this way coincides with the trivial lower bound; see Theorem 6.1.

(6) Fix $N_i > 0$, $i = 1, \dots, n$ and a partition $I_1, \dots, I_m$ of $\{1, \dots, n\}$, i.e., $I_1, \dots, I_m$ are nonempty pairwise disjoint subsets of $\{1, \dots, n\}$ whose union is all of $\{1, \dots, n\}$. Set $\mathbf{g} = (g_1, \dots, g_m)$ where $g_j := 1 - \sum_{i \in I_j} (x_i/N_i)^d$, $j = 1, \dots, m$. In this situation, condition $(**)$ holds and Theorem 4.2 (1) and (2) apply to produce lower bound for f on $K_{\mathbf{g}}$, where $K_{\mathbf{g}}$ is the product of hyperellipsoids defined by

$$K_{\mathbf{g}} := \{\mathbf{x} \in \mathbb{R}^n : \sum_{i \in I_j} (x_i/N_i)^d \leq 1, j = 1, \dots, m\}.$$

(7) Examples (3) and (4) can be seen as special cases of (6): If each I_j is singleton, (6) produces the lower bound for f on $\prod_{i=1}^n [-N_i, N_i]$ described in (4). If there is just one I_j, i.e., $m = 1$ and $I_1 = \{1, \dots, n\}$, and if $N_i = \sqrt[d]{M}$, $i = 1, \dots, n$, then (6) produces to the lower bound for f on B_M described in (3).

(8) Table 1 records average running time for computation of $s(f, \mathbf{g})$ for large examples (where computation of $f^{(d)}_{\mathrm{sos},\mathbf{g}}$ is not possible). Here $\mathbf{g} = (g_1, \dots, g_m)$, $g_j = 1 - \sum_{i \in I_j} x_i^d$, $j = 1, \dots, m$, where $\{I_1, \dots, I_m\}$ is a partition of $\{1, \dots, n\}$. The average is taken over 10 randomly chosen partitions $\{I_1, \dots, I_m\}$ and polynomials f, each f having t terms and $\deg(f) \leq d$ with coefficients chosen from $[-10, 10]$. See also [3, Table 2] and [5, Table 3].

TABLE 1. Average runtime for computation of $s(f, \mathbf{g})$ (seconds) for various n, d and t.

n	$d \backslash t$	50	100	150	200
10	20	3.330	23.761	79.369	170.521
	40	5.730	43.594	159.282	421.497
	60	6.524	68.625	191.126	531.442
20	20	8.364	63.198	193.431	562.243
	40	16.353	149.137	473.805	1102.579
	60	31.774	304.065	782.967	1184.263
30	20	12.746	107.285	353.803	776.831
	40	46.592	310.228	753.356	1452.159
	60	58.838	539.738	1271.102	1134.887
40	20	15.995	148.827	423.117	989.318
	40	60.861	414.188	1493.461	1423.965
	60	95.384	784.039	1305.201	1093.932

(9) Table 2 records average running time for computation of $s(f, \mathbf{g})$ (the top number) and $f^{(d)}_{\mathrm{sos},\mathbf{g}}$ (the bottom number) for small examples. Here $\mathbf{g} = (g_1, \dots, g_m)$, $g_j = 1 - \sum_{i \in I_j} x_i^d$, $j = 1, \dots, m$, where $\{I_1, \dots, I_m\}$ is a partition of $\{1, \dots, n\}$. The average is taken over 10 randomly chosen partitions $\{I_1, \dots, I_m\}$ and polynomials

f, each f having t terms and $\deg(f) \leq d$ with coefficients chosen from $[-10, 10]$.[4] See also [3, Table 1] and [5, Table 2].

TABLE 2. Average runtime for computation of $s(f, \mathbf{g})$ and $f^{(d)}_{\mathrm{sos},\mathbf{g}}$ for various n, d and t.

n	$d\backslash t$	10	30	50	100	150	200	250	300	350	400
3	4	0.034	0.092								
		0.026	0.026								
	6	0.041	0.105								
		0.108	0.103								
	8	0.045	0.164								
		0.522	0.510								
4	4	0.042	0.116								
		0.045	0.054								
	6	0.043	0.133	0.285							
		0.662	0.632	0.624							
	8	0.053	0.191	0.446	2.479	7.089					
		12.045	13.019	11.956	12.091	12.307					
5	4	0.039	0.124	0.295							
		0.181	0.154	0.127							
	6	0.052	0.156	0.396	1.677	5.864	13.814				
		6.429	6.530	6.232	6.425	6.237	6.469				
	8	0.056	0.219	0.528	3.046	7.663	23.767	44.699	88.104	123.986	179.126
		340.321	243.860	225.746	205.621	222.619	220.887	224.018	218.994	219.085	213.348
6	4	0.043	0.140	0.340	1.545						
		0.453	0.422	0.422	0.448						
	6	0.053	0.194	0.429	2.079	7.138	16.212	34.464	71.465	102.342	166.345
		48.239	47.752	47.776	51.268	51.008	48.012	49.908	53.135	51.542	52.874
	8	0.066	0.251	0.681	3.389	11.985	36.495	74.088	148.113	174.163	269.544
		–	–	–	–	–	–	–	–	–	–

(10) Table 3 computes average values for the relative error

$$R = \frac{-s(-f, \mathbf{g}) - s(f, \mathbf{g})}{-(-f)^{(d)}_{\mathrm{sos},\mathbf{g}} - f^{(d)}_{\mathrm{sos},\mathbf{g}}}$$

for small examples. Here $\mathbf{g} = (g_1, \ldots, g_m)$, $g_j = 1 - \sum_{i \in I_j} x_i^d$, $j = 1, \ldots, m$, where $\{I_1, \ldots, I_m\}$ is a partition of $\{1, \ldots, n\}$. The average is taken over 20 randomly chosen partitions $\{I_1, \ldots, I_m\}$ and polynomials f, each f having t terms and $\deg(f) \leq d$ with coefficients chosen from $[-10, 10]$. Table 3 would seem to confirm that for fixed n, d the quality of the bound $s(f, \mathbf{g})$ is best when t is small, and for fixed d, t the quality of the bound $s(f, \mathbf{g})$ is best when n is large. Comparison of Table 3 with [3, Table 3] would seem to indicate that the quality of the bound $s(f, \mathbf{g})$ is best when $m = 1$.

[4]**Hardware and Software specifications.** Processor: Intel® Core™2 Duo CPU P8400 @ 2.26GHz, Memory: 3 GB, OS: UBUNTU 14.04-32 bit, SAGE-6.0

TABLE 3. Average values of R for various n, d and t.

n	$d\backslash t$	5	10	50	100	150	200	250	300	400
3	4	1.3688	1.6630							
	6	1.5883	2.0500	4.3726						
	8	2.0848	2.7636	5.6391	5.8140	6.9135				
4	4	1.2183	1.3420	3.3995						
	6	1.3816	2.9584	3.1116	4.6891	5.8067	6.5150			
	8	1.7630	2.2038	3.2685	4.4219	5.7929	7.0841	7.9489	8.7924	9.6068
5	4	1.2566	1.6701	3.1867	4.6035					
	6	2.3807	2.9424	3.3077	4.7939	5.6523	7.1996	8.6194	9.3317	10.134
	8	1.5557	1.9754	2.5204	3.9815	4.5404	5.1756	6.2214	6.6919	7.8921
6	4	1.2069	1.3876	3.0639	4.5326	4.9645	6.1414			
	6	1.2602	1.4854	2.8236	4.0256	4.5797	6.3479	6.9487	7.4866	8.8435
	8	1.0478	1.1616	2.4884	3.3896	4.0870	5.0809	5.6932	6.1994	10.567
7	4	1.1943	1.3360	2.8592	4.5334	5.5064	5.9837	7.4782	7.3568	
	6	1.2604	1.4529	2.6962	3.8035	4.5351	6.0305	6.1478	6.6746	9.5049
8	4	1.1699	1.4274	2.5942	4.0914	5.4111	6.2111	7.4479	8.3168	9.1369
	6	1.0454	1.1270	2.1316	3.2807	3.5468	4.8955	5.1211	5.4214	7.1872
9	4	1.2158	1.3214	2.9305	4.0624	5.9063	6.5985	7.9552	8.4233	11.810
10	4	1.1476	1.3441	2.2393	3.7136	5.7739	6.1791	6.3211	8.7773	10.428

5. VARIANTS OF THEOREM 4.2

We explain how the hypothesis of Theorem 4.2 can be weakened a bit when $m = 1$.

Theorem 5.1. *Suppose $m = 1$ and $(g_1)_{d,i} = 0 \Rightarrow f_{d,i} > 0$ for $i = 1, \dots, n$. Choose c so that $(g_1)_{d,i} < 0 \Rightarrow c \geq -\frac{f_{d,i}}{(g_1)_{d,i}}$ and $(g_1)_{d,i} > 0 \Rightarrow c > -\frac{f_{d,i}}{(g_1)_{d,i}}$, for each $i = 1, \dots, n$. Then (1) the hypothesis of Theorem 4.1 holds for $A := \begin{pmatrix} 1 & 0 \\ -c & 1 \end{pmatrix}$, and (2) if, in addition, $\Delta(-g_1) = \emptyset$, $g_1(0) \geq 0$, and $c > 0$, then $f^A_{\mathrm{gp},\mathbf{g}} = s(f, \mathbf{g})$.*

Proof. Since $A = \begin{pmatrix} 1 & 0 \\ -c & 1 \end{pmatrix}$, $\lambda_0 = \mu_0 = 1$, $\lambda_1 = -c + \mu_1$, and $G(\lambda) = f - \lambda_1 g_1 = f - (-c + \mu_1) g_1 = (f + c g_1) - \mu_1 g_1$, so $H(\mu) = -h_0 - \mu_1 h_1$ where $h_0 = -(f + c g_1)$ and $h_1 = g_1$. The hypothesis of Theorem 4.1 is that exactly one of $(h_0)_{d,i}, (h_1)_{d,i}$ is strictly negative for each $i = 1, \dots, n$. Since $(h_0)_{d,i} = -(f_{d,i} + c(g_1)_{d,i})$ and $(h_1)_{d,i} = (g_1)_{d,i}$ the proof of (1) is completely straightforward. The proof of (2) is similar to the proof of Theorem 4.2 (2), but it is a good deal simpler: If $\lambda_1 \in [0, \infty)$ and $\mathbf{z}$ is a feasible point of (4) then $(\mathbf{z}, \mathbf{w}, \mu_1)$, where $w_\alpha = |f_\alpha|$ and $\mu_1 = c + \lambda_1$, is

a feasible point of (3), and

$$H(\mu_1)(0) - \sum_{\alpha\in\Delta(f)^{<d}} (d-|\alpha|)\left[\left(\frac{\mathbf{w}_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)}$$
$$= G(\lambda_1)(0) - \sum_{\alpha\in\Delta(f)^{<d}} (d-|\alpha|)\left[\left(\frac{f_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)}.$$

It follows that $f^A_{\mathrm{gp},\mathbf{g}} \geq G(\lambda)_{\mathrm{gp}}$. □

Similarly, the hypothesis of Theorem 4.2 can be weakened when $m = 2$ and $f_{d,i} = 0$ for $i = 1, \ldots, n$.

Theorem 5.2. *Suppose $m = 2$, $f_{d,i} = 0$ for $i = 1, \ldots, n$ and $(g_2)_{d,i} = 0 \Rightarrow (g_1)_{d,i} < 0$ for $i = 1, \ldots, n$. Choose c so that $(g_2)_{d,i} < 0 \Rightarrow c \geq \frac{(g_1)_{d,i}}{(g_2)_{d,i}}$ and $(g_2)_{d,i} > 0 \Rightarrow c > \frac{(g_1)_{d,i}}{(g_2)_{d,i}}$, for each $i = 1, \ldots, n$. Then (1) the hypothesis of Theorem 4.1 holds for $A := \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -c & 1 \end{pmatrix}$, and (2) if, in addition, $\Delta(-g_j) = \emptyset$ for $j = 1, 2$, $g_k(0) \geq 0$, for $k = 1, 2$, and $c > 0$, then $f^A_{\mathrm{gp},\mathbf{g}} \geq \sup\{G(\lambda)_{\mathrm{gp}} : \lambda_1 > 0, \lambda_2 \geq 0\}$.*

Proof. Similar to the proof of Theorem 5.1. □

We also mention another variant of Theorem 4.2.

Theorem 5.3. *Assume (†): for each $i = 1, \ldots, n$, exactly one of the coefficients $(g_0)_{d,i}, \ldots, (g_m)_{d,i}$ is strictly negative. Then (1) Theorem 4.1 applies with $A := I$ (the identity matrix), (2) $f^I_{\mathrm{gp},\mathbf{g}} \leq s_0(f, \mathbf{g})$ and (3) if, in addition, $\Delta(-g_j) \cap \Delta(-g_k) = \emptyset$ for $0 \leq j < k \leq m$ and $g_k(0) \geq 0$ for $k = 1, \ldots, m$ then $f^I_{\mathrm{gp},\mathbf{g}} = s_0(f, \mathbf{g})$. Here, $s_0(f, \mathbf{g}) := \sup\{G(\lambda)_{\mathrm{gp}} : \lambda \in (0, \infty)^m\}$.*

Proof. (1) is clear. Arguing as in the proof of Theorem 4.1 we see that $f^I_{\mathrm{gp},\mathbf{g}} \leq f(0) - \rho$, where ρ is the optimum value of program (2) in Section 3, but where the unknowns $\lambda = (\lambda_1, \ldots, \lambda_m)$ are now required to satisfy the strict inequality $\lambda_j > 0$. (2) follows from this by a rather obvious modification of the proof of Theorem 3.1. The extra hypothesis in (3) implies that $f^I_{\mathrm{gp},\mathbf{g}} = f(0) - \rho = s_0(f, \mathbf{g})$. □

Observe that when $m = 1$ Theorem 5.1 provides a generalization of both Theorem 4.2 and Theorem 5.3. Ditto for Theorem 5.2 when $m = 2$ and $f_{d,i} = 0$ for $i = 1, \ldots, n$.

Observe also that, as was the case with Theorem 4.2, if $m = 1$, the hypothesis of Theorem 5.1 holds, $\Delta(-g_1) = \emptyset$, and A is chosen as in Theorem 5.1, or, if $m = 2$, the hypothesis of Theorem 5.2 holds, $\Delta(-g_j) = \emptyset$, $j = 1, 2$, and A is chosen as in Theorem 5.2, then program (3) reduces to program (5). Similarly, if the hypothesis of Theorem 5.3 holds, $\Delta(-g_j) \cap \Delta(-g_k) = \emptyset$ for $0 \leq j < k \leq m$, and $A = I$, then

$G(\lambda)_\alpha = \lambda_j (g_j)_\alpha$ for some j (depending on α), and program (3) reduces to the following one:

$$(6)\quad \begin{cases} \text{Minimize} & \sum_{j=1}^m \lambda_j g_j(0)^+ + \sum_{\alpha\in\Delta^{<d}} (d-|\alpha|)\left[\left(\frac{G(\lambda)_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)} \\ \text{s.t.} & \sum_{\alpha\in\Delta} z_{\alpha,i} \le G(\lambda)_{d,i}, \qquad i=1,\dots,n \\ & \left(\frac{\mathbf{z}_\alpha}{\alpha}\right)^\alpha = \left(\frac{G(\lambda)_\alpha}{d}\right)^d, \qquad \alpha\in\Delta^{=d} \end{cases}$$

where, for every $\alpha\in\Delta$, the unknowns $\mathbf{z}_\alpha = (z_{\alpha,i}) \in [0,\infty)^n$ satisfy $z_{\alpha,i}=0$ if and only if $\alpha_i = 0$, and the unknowns $\lambda = (\lambda_1,\dots,\lambda_m)$ satisfy $\lambda_j > 0$.

Remark 5.4. Theorems 4.2, 5.1, 5.2 and 5.3 can be applied in various cases:

(1) Assume there exist $j_1,\dots,j_\ell \in \{1,\dots,m\}$ such that for each $i=1,\dots,n$ there exists k such that $(g_{j_k})_{d,i} < 0$ and $(g_{j_p})_{d,i} = 0$ for $k < p \le \ell$. Here, $j_0 := 0$. Then one can apply Theorem 4.2 to compute a lower bound for f on $K_{(g_{j_1},\dots,g_{j_\ell})}$. Since $K_{\mathbf{g}} \subseteq K_{(g_{j_1},\dots,g_{j_\ell})}$, this is also a lower bound for f on $K_{\mathbf{g}}$.

(2) Assume there exist $j_1,\dots,j_\ell \in \{1,\dots,m\}$ such that for each $i=1,\dots,n$ exactly one of the coefficients $-f_{d,i}, (g_{j_1})_{d,i},\dots,(g_{j_\ell})_{d,i}$ is strictly negative. Then one can apply Theorem 5.3 to compute a lower bound for f on $K_{(g_{j_1},\dots,g_{j_\ell})}$. Since $K_{\mathbf{g}} \subseteq K_{(g_{j_1},\dots,g_{j_\ell})}$, this is also a lower bound for f on $K_{\mathbf{g}}$.

(3) If there exists $j\in\{1,\dots,m\}$ such that $(g_j)_{d,i} = 0 \Rightarrow f_{d,i} > 0$ for all $i=1,\dots,n$ then one can apply Theorem 5.1 to compute a lower bound for f on $K_{(g_j)}$. Since $K_{\mathbf{g}} \subseteq K_{(g_j)}$, this is also a lower bound for f on $K_{\mathbf{g}}$.

(4) If $f_{d,i} = 0$, $i=1,\dots,n$ and there exists $j,k\in\{1,\dots,m\}$, $j\ne k$ such that $(g_j)_{d,i} = 0 \Rightarrow (g_k)_{d,i} < 0$ for all $i=1,\dots,n$ then one can apply Theorem 5.2 to compute a lower bound for f on $K_{(g_j,g_k)}$. Since $K_{\mathbf{g}} \subseteq K_{(g_j,g_k)}$, this is also a lower bound for f on $K_{\mathbf{g}}$.

Example 5.5.

(1) Suppose $n=3$, $m=2$, $d=2$, $f = p+qx+ry+sz$, $g_1 = 1-x^2-y^2$, $g_2 = 1-z^2$. In this example, $K_{\mathbf{g}}$ is a cylinder and the lower bound for f on $K_{\mathbf{g}}$ obtained using Theorem 4.2 or Theorem 5.3 is $p-\sqrt{q^2+r^2}-|s|$, which is the exact minimum of f on $K_{\mathbf{g}}$.[5]

(2) Suppose $n=3$, $m=2$, $d=2$, $f = p+qx+ry+sz$, $g_1 = 2-x^2-y^2-z^2$, $g_2 = 1-z^2$. In this example, $K_{\mathbf{g}}$ is a sphere with polar caps removed and the lower bound for f on $K_{\mathbf{g}}$ obtained using Theorem 4.2 or Theorem 5.3 is

$$\begin{cases} p-\sqrt{q^2+r^2}-|s| & \text{if } s^2 \ge q^2+r^2 \\ p-\sqrt{2}\sqrt{q^2+r^2+s^2} & \text{if } s^2 \le q^2+r^2 \end{cases},$$

which is the exact minimum of f on $K_{\mathbf{g}}$.

(3) Suppose $n=2$, $m=2$, $d=2$, $f = p+qx+ry$, $g_1 = 1-2x^2+y^2$, $g_2 = 1+x^2-y^2$. In this example, Theorem 4.2 does not apply, and the lower

[5] In examples (1), (2) and (3) the geometric program is so small that it can be solved by hand.

bound for f on $K_{\mathbf{g}}$ obtained using Theorem 5.3 is $p - |q|\sqrt{2} - |r|\sqrt{3}$, which is the exact minimum of f on $K_{\mathbf{g}}$.

(4) Suppose $n = 2$, $m = 1$, $d = 4$.

i. If $f = 5x + 6y + x^3 - y^2$, $g_1 = 8 - xy - x^4 - y^4$, the lower bound obtained using Theorem 4.2 is -22.334, $f^{(d)}_{\mathrm{sos},\mathbf{g}} = -18.778$.

ii. If $f = 5x + 6y + x^3 - y^2 + 2xy$, $g_1 = 8 - x^4 - y^4 + x^2y^2$, the lower bound obtained using Theorem 4.2 is -31.815, $f^{(d)}_{\mathrm{sos},\mathbf{g}} = -20.588$.

iii. If $f = 5x + 6y + x^3 - y^2 + 2xy$, $g_1 = 8 + xy - x^4 - y^4 + x^2y^2$, the lower bound obtained using Theorem 4.2 is -31.815, $f^{(d)}_{\mathrm{sos},\mathbf{g}} = -23.247$.

In each of i, ii and iii, $f^{(d)}_{\mathrm{sos},\mathbf{g}}$ is the exact minimum of f on $K_{\mathbf{g}}$.

(5) Suppose $n = 3$, $m = 2$, $d = 4$, $f = x^4 + y^4 + z^4 - y^3 + xy$, $g_1 = 10x^3z + xyz^2 + z^2 - 1$, $g_2 = z^4 - x^2yz$. In this example, Theorem 4.2 and Theorem 5.2 do not apply, the lower bound obtained using Theorem 5.3 is $f^{I}_{\mathrm{gp},\mathbf{g}} = -0.485$, and $f^{(d)}_{\mathrm{sos},\mathbf{g}} = -0.468$. In this example, $f^{I}_{\mathrm{gp},\mathbf{g}} = f_{\mathrm{gp}} = -0.485$ is the exact minimum of f on $\mathbb{R}^3$ and $f^{(d)}_{\mathrm{sos},\mathbf{g}} = -0.468$ is the exact minimum of f on $K_{\mathbf{g}}$.

(6) Suppose $n = 2$, $m = 2$, $d = 4$.

i. If $f = x + y$, $g_1 = 1 - 2y + 6x^2 - x^4$, $g_2 = -x^3 - y^4$, the lower bound obtained using Theorem 4.2 is -4.64574, which is the exact minumum of f on $K_{\mathbf{g}}$.

ii. If $f = 7y - 2x^3$, $g_1 = y + 8y^2 + 2xy^2 - x^4$, $g_2 = -x^2y - y^4$, the lower bound obtained using Theorem 4.2 is -88.3437 and $f^{(d)}_{\mathrm{sos},\mathbf{g}} = -86.1157$, which is the exact minimum of f on $K_{\mathbf{g}}$.

(7) Suppose $n = 3$, $m = 1$, $d = 6$.

i. If $f = x + z^3 + y^6 + z^6$, $g_1 = 1 - x^6 + y^6$, then the lower bound obtained using Theorem 5.1 or Theorem 5.3 is -1.25, which is equal to the exact minimum of f on $K_{\mathbf{g}}$.

ii. If $f = x + z^3 + x^6 + y^6 + z^6$, $g_1 = 1 - x^6 + y^6$, then Theorem 5.1 applies with $A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, and $f^{A}_{\mathrm{gp},\mathbf{g}} = f_{*,\mathbf{g}} = -1.25$.

(8) Suppose $n = 2$, $m = 2$, $d = 6$, $f = -y - 2x^2$, $g_1 = y - x^4y + y^5 - x^6 - y^6$, $g_2 = y - 5x^2 + x^4y - x^6 - y^6$. The lower bound for f obtained using Theorem 4.2 is -3.593. Applying Theorem 4.1 with

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}$$

yields the better lower bound -2.652. The exact minimum of f on $K_{\mathbf{g}}$ is -1.0494.

6. The trivial bound on $\prod_{i=1}^n[-N_i, N_i]$

Fix $f \in \mathbb{R}[\mathbf{x}]$ and $\mathbf{N} = (N_1, \dots, N_n)$, $N_i > 0$, $i = 1, \dots, n$. If $\mathbf{x} \in \prod_{i=1}^n[-N_i, N_i]$, then

$$f(\mathbf{x}) = \sum f_\alpha \mathbf{x}^\alpha \geq f(\mathbf{0}) - \sum_{\alpha \in \Delta'(f)} |f_\alpha| \cdot |\mathbf{x}^\alpha| \geq f(\mathbf{0}) - \sum_{\alpha \in \Delta'(f)} |f_\alpha| \cdot \mathbf{N}^\alpha,$$

where $\Delta'(f) := \{\alpha \in \mathbb{N}^n : |\alpha| > 0 \text{ and } f_\alpha \mathbf{x}^\alpha \text{ is not a square in } \mathbb{R}[\mathbf{x}]\}$ and $\mathbf{N}^\alpha := \prod_{i=1}^n N_i^{\alpha_i}$. Set

$$f_{\mathrm{tr},\mathbf{N}} := f(\mathbf{0}) - \sum_{\alpha \in \Delta'(f)} |f_\alpha| \cdot \mathbf{N}^\alpha. \tag{7}$$

Thus $f_{\mathrm{tr},\mathbf{N}}$ is a lower bound for f on the hypercube $\prod_{i=1}^n[-N_i, N_i]$. We refer to $f_{\mathrm{tr},\mathbf{N}}$ as the *trivial bound* for f on $\prod_{i=1}^n[-N_i, N_i]$. If $N_i = \sqrt[d]{M}$, $i = 1, \dots, n$, this coincides with the trivial bound defined in [3, Section 3].

Suppose now that d is an even integer, $d \geq \max\{2, \deg f\}$. Define $\mathbf{g} = (N_1^d - x_1^d, \dots, N_n^d - x_n^d)$. We want to compare $s(f, \mathbf{g})$ with $f_{\mathrm{tr},\mathbf{N}}$.

Theorem 6.1. *Set-up as above. Then*

(1) $s(f, \mathbf{g}) \geq f_{\mathrm{tr},\mathbf{N}}$.

(2) If $f_{d,i} \leq 0$ *for* $i = 1, \dots, n$ *then* $s(f, \mathbf{g}) = f_{\mathrm{tr},\mathbf{N}}$. *In particular, if* $\deg f < d$ *then* $s(f, \mathbf{g}) = f_{\mathrm{tr},\mathbf{N}}$.

We remark that the hypothesis of Theorem 6.1 (2) is indeed necessary: If $n = 1$, $f = x^2 - x$, $d = 2$, $N_1 = 1$, then $f_{\mathrm{tr},\mathbf{N}} = -1$, $s(f, \mathbf{g}) = f_{\mathrm{gp}} = -\frac{1}{4}$.

Proof. By making the change of variables $y_i = \frac{x_i}{N_i}$, $i = 1, \dots, n$, we are reduced to the case where $N_1 = \cdots = N_n = 1$. By definition of $\mathbf{g}$,

$$G(\lambda) = f - \sum_{i=1}^n \lambda_i(1 - x_i^d) = f(\mathbf{0}) - \sum_{i=1}^n \lambda_i + \sum_{\alpha \in \Omega(f)} f_\alpha \mathbf{x}^\alpha + \sum_{i=1}^n (f_{d,i} + \lambda_i) x_i^d,$$

and $s(f, \mathbf{g})$ is obtained by maximizing the objective function

$$f(\mathbf{0}) - \sum_{i=1}^n \lambda_i - \sum_{\alpha \in \Delta(f)^{<d}} (d - |\alpha|) \left[\left(\frac{f_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha \right]^{1/(d-|\alpha|)} \tag{8}$$

subject to

$$\begin{cases} \sum\limits_{\alpha \in \Delta(f)} z_{\alpha,i} \leq f_{d,i} + \lambda_i, & i = 1, \dots, n \\ \left(\frac{f_\alpha}{d} \right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha} \right)^\alpha = 1, & \alpha \in \Delta(f)^{=d} \end{cases} \tag{9}$$

where, $\lambda_i \geq 0$, $z_{\alpha,i} \geq 0$ and $z_{\alpha,i} = 0$ if and only if $\alpha_i = 0$.

(1) Define $z_{\alpha,i} := \alpha_i \frac{|f_\alpha|}{d}$ and $\lambda_i := \max\{0, \sum_{\alpha \in \Delta(f)} z_{\alpha,i} - f_{d,i}\}$. One checks that, for this choice of $z_{\alpha,i}$ and λ_i, the constraints of (9) are satisfied. Observe

also that $f_{d,i} \geq 0 \Rightarrow \lambda_i \leq \sum_{\alpha\in\Delta(f)} z_{\alpha,i}$ and $f_{d,i} < 0 \Rightarrow \lambda_i = \sum_{\alpha\in\Delta(f)} z_{\alpha,i} - f_{d,i}$. Consequently,

$$\begin{aligned}\sum_{i=1}^{n}\lambda_i \leq & \sum_{f_{d,i}\geq 0}(\sum_{\alpha\in\Delta(f)} z_{\alpha,i}) + \sum_{f_{d,i}<0}(\sum_{\alpha\in\Delta(f)} z_{\alpha,i} - f_{d,i})\\ =&\sum_{i=1}^{n}(\sum_{\alpha\in\Delta(f)} z_{\alpha,i}) + \sum_{f_{d,i}<0}|f_{d,i}|\\ =& \sum_{\alpha\in\Delta(f)}(\sum_{i=1}^{n} z_{\alpha,i}) + \sum_{f_{d,i}<0}|f_{d,i}|\\ =& \sum_{\alpha\in\Delta(f)}|\alpha|\cdot\frac{|f_\alpha|}{d} + \sum_{f_{d,i}<0}|f_{d,i}|,\end{aligned}$$

and

$$\begin{aligned}s(f,\mathbf{g}) \geq & f(\mathbf{0}) - \sum_{i=1}^{n}\lambda_i - \sum_{\alpha\in\Delta(f)^{<d}}(d-|\alpha|)\left[\left(\frac{f_\alpha}{d}\right)^d\left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)}\\ \geq & f(\mathbf{0}) - \sum_{\alpha\in\Delta(f)}|\alpha|\cdot\frac{|f_\alpha|}{d} - \sum_{f_{d,i}<0}|f_{d,i}| - \sum_{\alpha\in\Delta(f)^{<d}}(d-|\alpha|)\cdot\frac{|f_\alpha|}{d}\\ =& f(\mathbf{0}) - \sum_{\alpha\in\Delta(f)}|f_\alpha| - \sum_{f_{d,i}<0}|f_{d,i}|\\ =& f_{\mathrm{tr},\mathbf{N}}.\end{aligned}$$

(2) Suppose $(\mathbf{z},\lambda)$ satisfies (9). Since we are trying to maximize (8), we may as well assume each λ_i is chosen as small as possible, i.e., $\lambda_i = \max\{0, \sum_{\alpha\in\Delta(f)} z_{\alpha,i} - f_{d,i}\}$. Since we are also assuming $f_{d,i} \leq 0$, this means $\lambda_i = \sum_{\alpha\in\Delta(f)} z_{\alpha,i} - f_{d,i}$. Then

$$\begin{aligned}& f(\mathbf{0}) - \sum_{i=1}^{n}\lambda_i - \sum_{\alpha\in\Delta(f)^{<d}}(d-|\alpha|)\left[\left(\frac{f_\alpha}{d}\right)^d\left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)}\\ &= f(\mathbf{0}) - \sum_{i=1}^{n}(\sum_{\alpha\in\Delta(f)} z_{\alpha,i} - f_{d,i}) - \sum_{\alpha\in\Delta(f)^{<d}}(d-|\alpha|)\left[\left(\frac{f_\alpha}{d}\right)^d\left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)}\\ &= f(\mathbf{0}) - \sum_{\alpha\in\Delta(f)^{<d}}(\sum_{i=1}^{n} z_{\alpha,i} + (d-|\alpha|)\left[\left(\frac{f_\alpha}{d}\right)^d\left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)})\\ &- \sum_{\alpha\in\Delta(f)^{=d}}(\sum_{i=1}^{n} z_{\alpha,i}) - \sum_{i=1}^{n}|f_{d,i}|.\end{aligned}$$

We *claim* that, for each $\alpha \in \Delta(f)^{<d}$, the minimum value of

$$\sum_{i=1}^{n} z_{\alpha,i} + (d-|\alpha|)\left[\left(\frac{f_\alpha}{d}\right)^d\left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha\right]^{1/(d-|\alpha|)}$$

subject to $z_{\alpha,i} \geq 0$ and $z_{\alpha,i} = 0$ iff $\alpha_i = 0$ is $|f_\alpha|$; and that, for each $\alpha \in \Delta(f)^{=d}$, the minimal value of

$$\sum_{i=1}^{n} z_{\alpha,i}$$

subject to $\left(\frac{f_\alpha}{d}\right)^d \left(\frac{\alpha}{\mathbf{z}_\alpha}\right)^\alpha = 1$, $z_{\alpha,i} \geq 0$ and $z_{\alpha,i} = 0$ iff $\alpha_i = 0$ is also equal to $|f_\alpha|$. It follows from the claim that the maximum value of (8) is equal to

$$s(f, \mathbf{g}) = f(\mathbf{0}) - \sum_{\alpha \in \Delta(f)} |f_\alpha| - \sum_{i=1}^{n} |f_{d,i}| = f_{\mathrm{tr},\mathbf{N}}.$$

In proving the claim, one can reduce first to the case where each α_i is strictly positive. The claim, in this case, is a consequence of the following lemma. □

Lemma 6.2. *Suppose $\alpha_i > 0$, $i = 1, \ldots, n$.*

(1) For $|\alpha| < d$, the minimum value of

$$\sum_{i=1}^{n} z_i + (d - |\alpha|) \left[(\frac{f_\alpha}{d})^d \prod_{i=1}^{n} \frac{\alpha_i^{\alpha_i}}{z_i^{\alpha_i}} \right]^{1/(d-|\alpha|)}$$

on the set $(0, \infty)^n$ is equal to $|f_\alpha|$. The minimum occurs at $z_i = \alpha_i \cdot \frac{|f_\alpha|}{d}$, $i = 1, \ldots, n$.

(2) For $|\alpha| = d$, the minimum value of $\sum_{i=1}^{n} z_i$ subject to $z_i > 0$ and $(\frac{|f_\alpha|}{d})^d \cdot \prod_{i=1}^{n} \frac{\alpha_i^{\alpha_i}}{z_i^{\alpha_i}} = 1$ is equal to $|f_\alpha|$. The minimum occurs at $z_i = \alpha_i \cdot \frac{|f_\alpha|}{d}$, $i = 1, \ldots, n$.

Proof. the optimization problem in (1) is equivalent to the problem of minimizing the function $\sum_{i=1}^{n+1} z_i$ subject to $z_i > 0$ and $(\frac{|f_\alpha|}{d})^d \cdot \prod_{i=1}^{n} \frac{\alpha_i^{\alpha_i}}{z_i^{\alpha_i}} \cdot \frac{(d-|\alpha|)^{d-|\alpha|}}{z_{n+1}^{d-|\alpha|}} = 1$. In this way, (1) reduces to (2). The proof of (2) is straightforward, e.g., making the change in variables $w_i = \frac{z_i d}{\alpha_i |f_\alpha|}$ we are reduced to minimizing $\sum_{i=1}^{n} \alpha_i w_i$ subject to $\prod_{i=1}^{n} w_i^{\alpha_i} = 1$. Using the relation between the arithmetic and geometric mean yields

$$\frac{\sum_{i=1}^{n} \alpha_i w_i}{d} \geq \sqrt[d]{\prod_{i=1}^{n} w_i^{\alpha_i}} = 1,$$

i.e., $\sum_{i=1}^{n} \alpha_i w_i \geq d$. On the other hand, if we take $w_i = 1$, then $\sum_{i=1}^{n} \alpha_i w_i = |\alpha| = d$. Thus the minimum occurs at $w_i = 1$, i.e., $z_i = \alpha_i \frac{|f_\alpha|}{d}$, $i = 1, \ldots, n$, and the minimum value of $\sum_{i=1}^{n} z_i$ is $\sum_{i=1}^{n} \alpha_i \frac{|f_\alpha|}{d} = |\alpha| \frac{|f_\alpha|}{d} = |f_\alpha|$. □

Remark 6.3.

(1) Suppose $I_1, \ldots, I_\ell$ and $J_1, \ldots, J_m$ are partitions of $\{1, \ldots, n\}$ with $I_1, \ldots, I_\ell$ finer than $J_1, \ldots, J_m$,

$$G(\lambda) = f - \sum_{p=1}^{\ell} \lambda_p \left(1 - \sum_{i \in I_p} (\frac{x_i}{N_i})^d \right), \; H(\mu) = f - \sum_{q=1}^{m} \mu_q \left(1 - \sum_{i \in J_q} (\frac{x_i}{N_i})^d \right).$$

One checks that if $\mu_q = \sum_{I_p \subseteq J_q} \lambda_p$, then $G(\lambda)_{\mathrm{gp}} \leq H(\mu)_{\mathrm{gp}}$. It follows that $s(f, \mathbf{g}) \leq s(f, \mathbf{h})$ where

$$\mathbf{g} = \left(1 - \sum_{i \in I_1} (\frac{x_i}{N_i})^d, \ldots, 1 - \sum_{i \in I_\ell} (\frac{x_i}{N_i})^d\right), \quad \mathbf{h} = \left(1 - \sum_{i \in J_1} (\frac{x_i}{N_i})^d, \ldots, 1 - \sum_{i \in J_m} (\frac{x_i}{N_i})^d\right).$$

(2) Similarly, one checks that if

$$H(\lambda) = f - \lambda(1 - \sum_{i=1}^{n} (\frac{x_i}{N_i})^d), \; I(\mu) = f - \sum_{j=1}^{n} \mu_j (\frac{1}{n} - (\frac{x_i}{N_i})^d)$$

where $\mu_j = \lambda$, $j = 1, \ldots, n$, then $H(\lambda) = I(\mu)$. It follows that $s(f, \mathbf{h}) \leq s(f, \mathbf{i})$ where

$$\mathbf{h} = \left(1 - \sum_{i=1}^{n} (\frac{x_i}{N_i})^d\right), \; \mathbf{i} = \left(\frac{1}{n} - (\frac{x_1}{N_1})^d, \ldots, \frac{1}{n} - (\frac{x_n}{N_n})^d\right).$$

(3) In particular, (1) and (2) imply

$$s(f, \mathbf{g}) \leq s(f, \mathbf{h}) \leq s(f, \mathbf{i}),$$

with $\mathbf{g}$ as in Theorem 6.1, $\mathbf{h}$ and $\mathbf{i}$ as in (2). Observe also that, by Theorem 6.1, $f_{\mathrm{tr},\mathbf{N}} \leq s(f, \mathbf{g})$ and $f_{\mathrm{tr},\mathbf{N}/\sqrt[d]{n}} \leq s(f, \mathbf{i})$ with equality holding if $f_{d,i} \leq 0$, $i = 1, \ldots, n$. This clarifies to some extent an observation made in [3, Section 3].

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